

AGRISMART AI

Smart decisions for healthier crops

Precision foliar disease diagnostics and climate-resilient agronomic advisory for Indian farmers.

SIH 2026	CORE AI	BONUS MODULES
Problem Statement 1	Crop disease detection	Irrigation • Weather • Sustainability • GenAI

PROJECT OVERVIEW

AgriSmart AI is a responsive web-based agricultural decision-support platform. Its mandatory SIH capability is computer-vision crop disease classification from a leaf image. It extends that core with weather intelligence, smart irrigation, a transparent sustainability indicator, and a grounded multilingual farmer assistant.

Verified development evaluation: 98.33% Accuracy and 98.33% Macro-F1 on a genuine PlantVillage held-out evaluation set.

Important: The official organizer-provided field-condition SIH test set was not available in the development environment. The 98.33% result is not claimed as the official SIH field-test score.

GitHub: github.com/VedPatel41/AgriSmart-AI
main • commit 77d17e2

1. SIH CORE — MODEL REPORT

Task

Crop-disease image classification into 15 trained classes, including healthy foliage, returning class label, confidence and precautionary advisory.

Dataset & Split

Genuine PlantVillage leaf images from the HuggingFace Parquet subset. Available pool: 11,322 training images and 2,916 test images. Reported development evaluation used 2,250 balanced training images (150/class) and 600 balanced held-out evaluation images (40/class).

Model / Approach

Pretrained EfficientNet-B0 using torchvision ImageNet-default weights. A 1280-feature backbone feeds a 15-class linear classifier. Input: 224×224 RGB; ImageNet normalization. Training: CrossEntropyLoss, Adam, learning rate 1×10^{-3} , 20 epochs, deterministic seed 42.

98.33%	98.33%	98.34%	98.33%
Accuracy	Macro-F1	Macro Precision	Macro Recall

Confusion Matrix & Per-Class Metrics

Complete confusion matrix and per-class precision/recall/F1 are stored in model/metrics.json and report/model_metrics.json. This report does not invent or reconstruct the matrix values.

Baseline & Limitations

No verified SIH organizer field-test baseline was available in the environment, so no numerical official-baseline comparison is claimed. PlantVillage imagery is controlled/lab-like; real field photographs may reduce confidence.

Prediction interface: python model/predict.py --image <path> and Flask /predict.

2. MODEL TAXONOMY & PLATFORM

#	Class	Type
1	Apple__Apple_scab	Disease
2	Apple__Black_rot	Disease
3	Apple__Cedar_apple_rust	Disease
4	Apple__healthy	Healthy
5	Blueberry__healthy	Healthy
6	Cherry_(including_sour)__Powdery_mildew	Disease
7	Cherry_(including_sour)__healthy	Healthy
8	Corn_(maize)__Cercospora_leaf_spot Gray_leaf_spot	Disease
9	Corn_(maize)__Common_rust_	Disease
10	Corn_(maize)__Northern_Leaf_Blight	Disease
11	Corn_(maize)__healthy	Healthy
12	Grape__Black_rot	Disease
13	Grape__Esca_(Black_Measles)	Disease
14	Grape__Leaf_blight_(Isariopsis_Leaf_Spot)	Disease
15	Grape__healthy	Healthy

Platform modules

Capability	Current implementation	SIH role
Disease detection	Leaf upload → EfficientNet-B0 → class + confidence + advisory	Core
Smart irrigation	Soil moisture + 24h rain probability + crop type + growth stage → rule-based recommendation	Bonus B
Weather intelligence	OpenWeatherMap current/forecast data, rain probability and foliar risk	Bonus C
Sustainability	0–100 water stewardship & microclimate adaptation index using transparent 60/40 rule	Bonus D
GenAI assistant	Grounded Gemini context using diagnosis, weather, irrigation and sustainability; multilingual prompts	Bonus E
Not implemented	Crop recommendation, IoT integration, autonomous agentic advisor	A / F / G

3. SYSTEM ARCHITECTURE & DATA FLOW

Layer	Flow
Farmer / Browser	Leaf image, soil moisture, location, crop and growth stage inputs
Flask REST API :5000	Validation, routing and JSON responses
Core AI	POST /predict → PyTorch EfficientNet-B0 → disease + confidence + advisory
Weather	GET/POST /weather → OpenWeatherMap → 24h rain probability + foliar risk
Irrigation	POST /irrigation → crop/stage-aware rule engine → Irrigate / Delay / Monitor
Sustainability	POST /sustainability → 60/40 stewardship formula → score + rating + suggestion
Assistant	POST /assistant → grounded JSON context → Google Gemini → multilingual farmer advice

Smart irrigation rules

Delay irrigation when next-24h rain probability is above 60%. Irrigate when soil moisture is below the effective crop/stage threshold and rain probability is below 30%; otherwise monitor. The effective baseline threshold is 30%, adjusted for high-water crops, reproductive stages, drought-hardy crops and maturity.

Sustainability formula

Score = $0.60 \times \text{Water Efficiency} + 0.40 \times \text{Weather Adaptation}$. Categories: ≥ 80 Excellent; 60–79 Good; 40–59 Fair; < 40 Needs Improvement. It is an indicative decision-support heuristic, not a certified regulatory carbon audit.

4. TESTING, SECURITY & REPRODUCIBILITY

Suite	Result
Frontend & architectural invariants	177/177 PASSED
Live API integration	12/12 PASSED
Final hardening & QA	8/8 PASSED
Smart irrigation	18/18 PASSED
Weather intelligence	10/10 PASSED
Sustainability scoring	9/9 PASSED
Gemini guardrails/error tests	10/10 PASSED
Assistant + E2E	17/17 PASSED

Technology stack

Frontend: HTML5, CSS3, Vanilla ES6 JavaScript.

Backend: Python 3.10+, Flask 3.0+, Flask-CORS, Werkzeug.

ML: PyTorch 2.1+, Torchvision EfficientNet-B0.

APIs: OpenWeatherMap REST API and Google Gemini REST API.

Security

External API credentials are supplied through environment variables. The tracked .env.example is sanitized to placeholders. The current implementation uses port 5000 and the final model checkpoint is 16.4 MB.

Reproducibility

Install dependencies from requirements.txt, configure environment variables, run the Flask backend, and use the documented CLI or /predict endpoint for single-image inference. The model checkpoint and metadata are committed to the GitHub repository.

GitHub

Repository: github.com/VedPatel41/AgriSmart-AI

Branch: main

Commit: 77d17e2 — Replace dummy model with SIH PlantVillage EfficientNet model

5. SIH ALIGNMENT, IMPACT & LIMITATIONS

Requirement	Current status
Core crop-disease detection	Implemented with real trained EfficientNet-B0 and 15-class prediction.
Honest development evaluation	PlantVillage held-out evaluation explicitly labeled; official field test unavailable.
Macro-F1 / confusion matrix / per-class metrics	98.33% Macro-F1; detailed metrics and matrix stored in model/metrics.json.
Prediction interface	CLI and Flask /predict verified.
Farmer-facing result/advisory	Implemented for all 15 classes.
Bonus B / C / D / E	Implemented: irrigation, weather intelligence, sustainability, grounded GenAI assistant.

Impact

AgriSmart AI connects visual crop diagnosis with water-aware irrigation, weather-driven risk context, transparent stewardship scoring and farmer-friendly explanations. The goal is earlier awareness, reduced unnecessary irrigation and more accessible agricultural decision support.

Honest limitations

- PlantVillage imagery is controlled compared with real farm photographs; field generalization is not established by the 98.33% result.
- Official organizer-provided SIH field-condition test data was not available, so the official judging benchmark remains pending.
- Diagnostics, irrigation alerts and sustainability scores are decision-support aids; treatment decisions should be cross-checked with qualified local agricultural extension/KVK specialists.
- Hyper-local weather accuracy depends on reporting-station availability and proximity.
- Weather and GenAI features depend on external APIs and valid configuration.

OFFICIAL FIELD-TEST STATUS

Official organizer-provided field-condition test data was not available in the current development environment; final official field-test evaluation remains pending.

The measured 98.33% Macro-F1 was obtained on the genuine PlantVillage held-out validation set and is not represented as the official SIH field test.

AgriSmart AI

Intelligent Agriculture for a Sustainable Future